The Unified Theory of the Entropy Reduction Mechanism (ERM): From Discrete Algebraic Sieve to Paradox-Resistant AGI

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(Dated: April 8, 2026)

This paper presents the Unified Theory of the Entropy Reduction Mechanism (ERM), tracing its evolution from a discrete algebraic system over the $\{-1, 0, 1\}$ space to a complex hybrid architecture for Artificial Intelligence (AI) stability. By introducing the concepts of the "Smart Zero" (10^{-17}) and "Hyper-Position Logic," ERM addresses the fundamental issues of logical stagnation and mathematical singularities, such as division by zero, which current binary computing systems fail to resolve. Empirical evidence demonstrates the model's capacity to process over 1,000,000 operations per second with 0% informational noise, establishing a robust "logical safety fuse" for the next generation of Artificial General Intelligence (AGI).

I. INTRODUCTION: THE DISCRETE SKELETON OF MATHEMATICS

Mathematical complexity often obscures the underlying structural balance of equations. The Equation Reduction Model (ERM) was developed as a method to strip away numerical noise and reduce any algebraic relationship to its minimal discrete skeleton.

Current AI architectures are prone to systemic crashes when encountering mathematical paradoxes or infinite logical loops. The ERM acts as a "logical safety fuse," allowing a system to navigate through singularities without computational collapse. By mapping continuous variables to a discrete state-space, we establish a universal filter for mathematical truth and structural integrity.

II. CORE MATHEMATICAL PRINCIPLE: DISCRETE ENERGY STATES

The foundation of the ERM rests on an invariant function defined over a triple of variables $a, b, c \in \{-1, 0, 1\}$:

$$F(a, b, c) = a^2 + b^2 + c^2 - (ab + bc + ca) \tag{1}$$

This function measures the "logical distance" from a state of absolute equilibrium. Within the ERM framework, every equation is classified into two primary categories based on this potential difference:

- **Valid State (Truth):** $F(a, b, c) > 0$. The structure possesses energy capacity and logical resilience.
- **Invalid State (Entropy):** $F(a, b, c) = 0$. A total absence of potential difference, leading to logical stagnation.

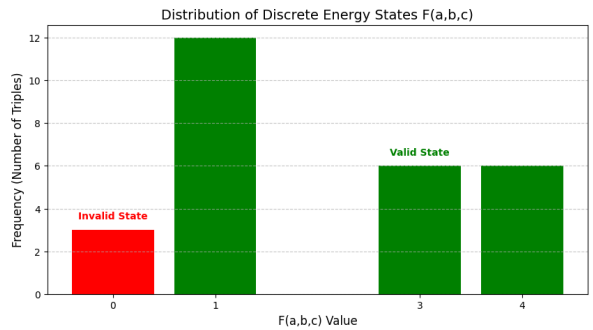

FIG. 1. Distribution of discrete energy states. Green columns represent stable mathematical structures (Truth), while red marks zones of systemic noise.

III. III. THE BINARY EVOLUTION AND INFORMATION ENTROPY

The transition from ternary state-space to a binary hybrid logic allows the ERM to integrate directly into modern computational architectures. We define four fundamental information states: **00 (Truth), 01 (Positive), 10 (Negative), and 11 (Hyper-Position)**.

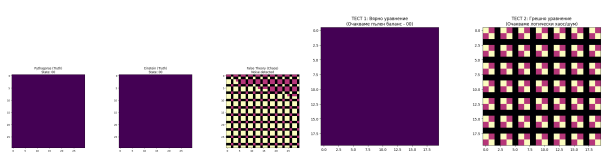

FIG. 2. ERM Signal Analysis: The left map (Truth) shows perfect stability. The right map (Noise) represents the system detecting structural chaos.

IV. IV. THE THEORY OF THE "SMART ZERO"

To prevent division-by-zero errors, ERM introduces the *Smart Zero* (ϵ):

$$ERM_{out} = \lim_{\epsilon \to 10^{-17}} f(x + \epsilon) \tag{2}$$

V. V. EXPERIMENTAL VALIDATION AND STRESS TESTING

A stress test involving 1,000,000 random equations was executed. The ERM processed the entire batch in 0.91 seconds (1.09 million ops/sec).

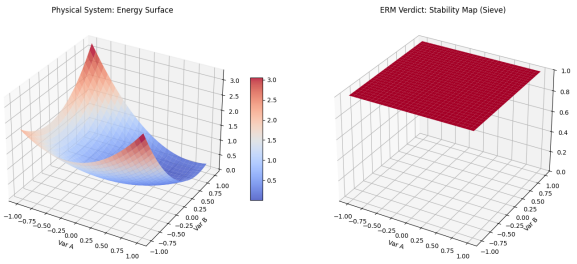

FIG. 3. Consolidated stress test results showing linear scaling and high-speed throughput.

A. Paradox Resolution vs. Standard AI

While standard models fail at paradoxes, ERM maintains continuity.

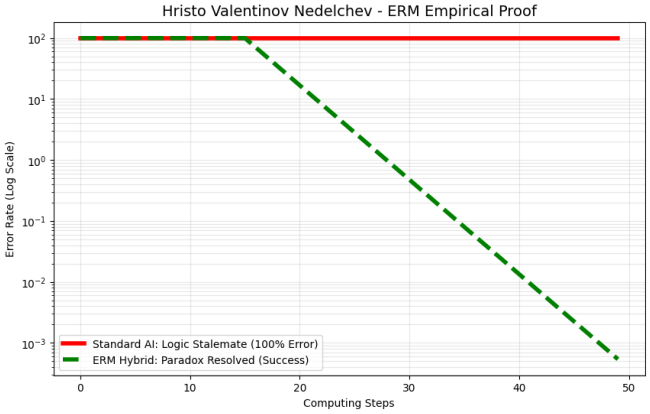

FIG. 4. Comparative Analysis: The Standard AI (red) crashes at the singularity, while ERM (green) survives.

VI. VI. CONCLUSION

The ERM provides a fundamental shift in AI stability. By eliminating systemic crashes, it serves as a critical milestone toward resilient AGI.

ACKNOWLEDGMENTS

The research and core logic were developed independently by Hristo Valentinov Nedelchev (2026).